

Clustering Methodes

1. Partitioning Methods

Method	When to Use	Pros	Cons
K-Means	Small-to-medium datasets where clusters are spherical and distinct.	•Efficient: $O(kn)$, faster than most other methods. •Simple: Easy to implement and understand.	•Spherical Only: Fails on non-convex shapes. •Outliers: Very sensitive to noise and outliers. •Parameters: Must guess k (number of clusters).
PAM (K-Medoids)	Small datasets where robustness to noise is critical.	•Robust: Uses real data points (medoids) as centers, making it less sensitive to outliers than K-Means.	•Slow: Does not scale well for large data. Complexity is high ($O(k(n-k)^2)$).
CLARA	Large datasets where you need a faster approximation of K-Medoids.	•Scalable: Handles larger data than PAM by using sampling. •Complexity: $O(ks^2 + k(n-k))$.	•Bias: Quality depends on sample size. If the best medoid isn't in the sample, results suffer.
CLARANS	Large spatial databases where you want better accuracy than CLARA.	•Quality: Does not confine search to a fixed sample; finds better clusters than CLARA.	•Speed: Slower than CLARA (Complexity is $O(n^2)$ in worst case).

2. Hierarchical Methods

Method	When to Use	Pros	Cons
AGNES (Agglomerative)	Small datasets where you need to see the hierarchy of relationships ¹⁶ .	• Hierarchy: Produces a dendrogram showing how clusters merge	• Rigid: Cannot undo a merge step once done¹⁹.

		17. • Flexible: Can define different "linkage" rules (single, complete, average) ¹⁸ .	• Slow: Complexity is at least $O(n^2)$ ²⁰ .
DIANA (Divisive)	Small datasets where a top-down approach is preferred ²¹ .	• Hierarchy: Provides the inverse order of AGNES ²² .	• Expensive: Computationally heavy to decide where to split clusters ²³ .
BIRCH	Big Data (numerical only) that does not fit in memory ^{24,24,24,24} .	• Fast: Finds clusters in a single scan of the database ²⁵ . • I/O Efficient: Minimizes disk scans ²⁶ .	• Numeric Only: Cannot handle categorical data ²⁷ . • Spherical: Tends to find only spherical clusters ²⁸ .
CHAMELEON	Complex datasets with arbitrary shapes and varying densities ^{29,29,29} .	• High Quality: Handles arbitrary shapes/densities better than BIRCH or DBSCAN ³⁰ . • Dynamic: Models clusters based on interconnectivity and closeness ³¹ .	• Slow: High complexity ($O(n^2)$) ³² . • Complex: Two-phase algorithm (Graph partitioning + Hierarchical) ³³ .

3. Density-Based Methods

Method	When to Use	Pros	Cons
DBSCAN	Spatial data with noise and arbitrary shapes ³⁴ .	• Shapes: Finds non-spherical (e.g., "S" shaped) clusters ³⁵ .	• Varying Density: Fails if clusters have different densities ³⁸ .

		• No \$k\$: No need to specify number of clusters 36. • Noise: Handles outliers well 37.	• Parameters: Sensitive to <i>Eps</i> and <i>MinPts</i> settings 39.
OPTICS	Data with varying densities where DBSCAN fails 40.	• Structure: Produces a "cluster-ordering" to visualize structure 41. • Flexible: Handles varying densities better than DBSCAN 42.	• Slow: Slower than DBSCAN; output requires interpretation (plot analysis) 43.
DENCLUE	Noisy data where a strong mathematical description is needed 44.	• Math Foundation: Uses solid statistical density functions 45. • Fast: Faster than DBSCAN (up to 45x) 46.	• Tuning: Needs accurate choice of parameters σ and ξ 47.

4. Grid-Based Methods

Method	When to Use	Pros	Cons
STING	Spatial data queries (e.g., GIS).	• Very Fast: Query processing is $O(g)$ (depends on grid size, not data size). • Parallel: Easy to parallelize.	• Rigid: Cluster boundaries are horizontal/vertical only (no diagonals). • Accuracy: Depends on the granularity of the grid.
CLIQUE	High-dimensional data (Subspace Clustering).	• Subspaces: Automatically finds clusters in subsets of dimensions.	• Accuracy: Tuning grid size and density threshold is difficult;

		• Scalable: Scales linearly with input size.	simplicity reduces accuracy.
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